2012 APSMO Set 3

1 Problems

Problem 1.1 A whole number is a divisor of a second whole number if the first divides the second without remainder.

How many different divisors does 120 have?

Problem 1.2 A work crew of 30 people can build a concrete wall in 8 days.

How long would it take a work crew of 16 people to build the same wall if each person in both crews work at the same rate?

Problem 1.3 In the multiplication problem below, the letters A, B, C, D represent different digits.

What is the sum of A and C?

$$\begin{array}{r} A\,B \\ \times\ C\,D \\ \hline 5\,7 \\ 7\,6 \\ \hline 8\,1\,7 \end{array}$$

Problem 1.4 On the table there is a set of six coins consisting of three 1 cent coins, two 5 cents coins and one 10 cents coin.

How many different amounts of money may be made by using one or more coins from this set?

Problem 1.5 The pages of a book are numbered from 1 to 200.

What is the total number of times that the digit “1” appears on these pages?